Cognitive Skills Acquired From Video Games

Emma G. Cunningham, University of Wisconsin-Madison and C. Shawn Green, University of Wisconsin-Madison

https://doi.org/10.1093/acrefore/9780190228613.013.1468
Published online: 20 September 2023

Summary

Due to the massive engagement with video games worldwide in innumerable forms and iterations, researchers have sought to understand the impact playing video games might have on the human brain and behavior. Although research on video games resides in a vast array of disciplines, including social, developmental, clinical, and educational psychology, this work focuses on research specifically in the cognitive sphere. From early research providing sound evidence for the positive impacts of action games on perceptual cognitive skills, to recent work refining methodologies for differentiating the effects of a wide range of embedded mechanics within broader game genres, the field has addressed a number of increasingly complex and critical questions. Research in the field has explored the effects of many game genres' unique mechanics and in-game goals. Specifically, studies have found that action games positively impact perceptual skills as well as higher-order attentional control and executive function skills, while game genres that utilize action-oriented mechanics including Action-Role Playing Games and Real Time Strategy games also induce similar effects, if to a lesser extent. These results have been observed both through correlational studies, where player status is an existing characteristic of participants, and through intervention studies, where novice participants are trained on a specific game to establish causality between game play and cognitive performance. Although less research has been dedicated to the effects of puzzle games, playing such games has been found to impact higher cognitive skills such as problem-solving and fluid intelligence. Building upon this body of work, future research should explore the cognitive impacts of a more diverse set of game types, in-game experiences, and cognitive constructs as well as the mechanisms through which they are impacted. This should include work dedicated to the effects of puzzle and mini games, and the impact of games on higher cognitive skills including planning, problem-solving, and fluid intelligence, where relatively little research has been dedicated in the past. Further, research should explore the differences in training outcomes from games, between immediate transfer of skills from training to test and the enhancement of the meta-skill of "learning to learn." Together, such work will allow game play to continue to evolve from pure entertainment to a force for good.

Keywords: video games, action video games, real time strategy games, role playing games, puzzle games, cognitive skills, cognitive training, perceptual skills, perceptual training, cognitive enhancement

Subjects: Communication and Technology, Communication and Culture

Why Video Games?

The popularity of video games cannot be overstated. The first recognizable at-home video game system, the Magnavox Odyssey, was released in 1972 and sold around 300,000 units. Approximately 30 years later, the so-called "7th generation" consoles (e.g., XBOX 360, PlayStation 3, Nintendo Wii) sold around 300 million units. And finally, fast-forwarding to the

2020s, where games can easily be played on phones, tablets, even on the back of airplane headrests, estimates suggest there are somewhere around three billion video gamers worldwide (Statista, 2023). Critically, video games long ago left the pure "entertainment" sphere and instead have come to permeate myriad distinct areas of society, including workplaces (e.g., training games and simulations), schools (e.g., "edu"-games), and athletics (e.g., "eSports").

Throughout recent human history, whenever a form of media or technology has reached such levels of saturation among the general population, questions have nearly always arisen regarding the potential impact of the form of media or technology on the human experience. Very frequently, these questions grew out of implicit fears that any changes induced by the given form of technology would be "for the worse." These include, for instance, concerns that the phonograph or radio might reduce literacy rates or that violence in television or movies would spur a national crime wave.

It is thus not surprising that research focused on the potential impact of video game play has massively accelerated over the past 50 years. For instance, from 1972 to 1982, the PubMed database of life sciences research returns fewer than one published research paper per year under the label "video games"; from 2012 to 2022, this same search results in nearly 900 papers indexed per year. Given the complex and multifaceted nature of video games, research on their potential impact spans essentially all of the behavioral sciences—from social psychology (e.g., research focused on aggression and violence as well as on prosocial behaviors and empathy), to developmental psychology (e.g., on how children's development is impacted by video games), to clinical psychology (e.g., in potential reciprocal relationships between video game play and various mental health disorders), to educational psychology (e.g., in the potential use of games in the classroom), to the focus of this article—the relationship between video game play and cognitive skills. Interestingly, while there are pockets of research that have examined potential negative impacts of gaming on certain aspects of cognitive skill, a sizable portion of the work to date has focused on the potential for video games to improve, rather than degrade, human cognitive function. All told, the existing combination of basic science and translational research suggests that it may be possible to purposefully harness video games to ensure that any changes, at least with respect to cognitive function, are "for the better."

Why Should There Be a Relationship Between Cognitive Function and Video Game Play?

While there was perhaps some initial tendency for video games to be viewed as "mindless entertainment" (indeed, the senior author's very first conference poster was entitled, "Video Game Playing: Rot Your Brain or Expand Your Mind?" which was not, at that time, viewed as a strawman), there was nonetheless very early recognition that video games, in a way that was unlike other forms of media, could be exceedingly perceptually, cognitively, and motorically challenging. This can be seen, for instance, in work as far back as the 1980s, in which the U.S. military funded research that explored the potential use of video games specifically as tests of cognitive and motor performance (Dorval & Pépin, 1986; Greenfield, Brannon, & Lohr, 1994;

Vernon et al., 1985). In the same vein, research at that time also noted the similarity between the processing demands inherent in some video games and the demands of tasks believed to be highly cognitively demanding (e.g., as related to piloting and landing aircraft).

This understanding that video games could place demand, or load, upon particular cognitive systems forms the core of many theories seeking to explain why video game play might impact those systems (Bowman, 2020). In short, it has been argued that when cognitive skills are placed under load, the brain responds by strengthening the requisite skills so as to perform better in the next instance when strain is placed on the system. And while there are definitely significant issues with the "brain as muscle" metaphor, there are obvious similarities here. Repeated strain on a muscle indicates to the body that this muscle should be reinforced in the event that a similar amount of strain will be placed on it again. Cognitive skills are argued to behave much the same; the level and frequency of load placed on a cognitive skill determines the amount by which the skill will be strengthened.

This exact reasoning can be seen very clearly, for example, in early and seminal research by Greenfield, DeWinstanley, et al. (1994), where they stated,

> The hypothesis [that gaming would enhance divided visual attention] came from the observation that in most video games, a player must be able to deal with events occurring simultaneously at several locations on the video screen. . . . It was therefore hypothesized that: (a) video game experts would be better than nonexperts at tasks requiring divided visual attention, and (b) video game experience would play a causal role in improving performance on such tasks. (p. 108)

Similar reasoning regarding the potential impact of various types of games, then, continues in much the same vein in recent research endeavors in the field.

What Types of Games Have Been Explored for Their Possible Cognitive Impacts?

Although the popular media tends to frame questions in this space as, "How does 'video gaming' alter cognitive function?" once the perspective that the impact of games comes from the particular challenges inherent in games is taken instead, it should immediately become clear that "video game" is too high-level of a category label to be of use scientifically. Indeed, within the modern video game industry, there exists an immense amount of variety of game experiences that challenge different cognitive functions and do so to different degrees. It should therefore follow that these varied types of games should impact cognition in diverse ways. Therefore, the idea that "not all video games are created equal with respect to their impact on cognitive function" is perhaps the most important organizing principle of the literature in the early 21st century.

Consistent with this organizing principle, the types of games that have been examined by cognitive scientists reflect systematic differences in the cognitive functions that are believed to be challenged by the given game types. Arguably the most-studied game type in the cognitive area is so-called “action video games.” These games, which historically have comprised first-person and third-person shooter games, place extreme demands on a wide variety of subdomains of cognition. Such games typically involve the need to keep track of multiple, quickly moving objects. These objects are frequently somewhat transient (i.e., they pop in and out of view), it is necessary to constantly and efficiently switch between a more focused attentional state (e.g., when aiming at an enemy) and a more diffuse attentional state (e.g., when monitoring the full field of view for danger), and responses must be both fast and accurate. These games have therefore been argued to place load on basic perceptual skills, speed of processing, the capacity and fidelity of visual attention, task-switching, multitasking, executive functions, and spatial cognitive skills, among others, in the cognitive domain. In fact, a prominent theory in the literature posits that the highly stimulating in-game environments and cognitively taxing, varied, and time-constrained objectives impact a player’s general capacity for learning, thus allowing cognitive improvements gained during gameplay to transfer to unrelated non-game activities or tasks (see Bavelier et al., 2012 for review).

The exploration of action games, starting in the late 1990s and early 2000s, then inspired a plethora of research on many other types of games such as Real Time Strategy (RTS) games, which, while quick in-game decisions are necessary, do not contain the same level of fast-paced actions or reactions as is the case in action video games. More recently, games with RTS components often also include components previously associated only with action games such as the requirement to rapidly survey an environment for potential dangers and react appropriately (see Dale & Green, 2017b for a discussion of the evolution of game genres).

In addition to RTS games, Role Playing Games also increasingly include action game components which can therefore tap into well-studied areas of cognition. Examples of so-called Action-Role Playing Games are titles such as *Mass Effect* and *Skyrim*. In these games, players control one or more characters and work within that structure to take on obstacles and enemies while on a journey or quest of some sort. Critically, though, while the “role-playing” aspect is what gives games from this genre their main identity, these games also typically use combat mechanics that are more or less identical to classic first- and/or third-person shooter games. Research on this game type is then an excellent example of the predictions regarding the cognitive consequences of playing certain games which arise from the challenges that are inherent in the games. Although a game like *Mass Effect* would often be referred to as a “Role-Playing Game,” because the challenges are in many ways identical to those seen in “third-person shooter games,” the predicted cognitive impact would also be equivalent. In other words, for scientists in this field, game genre labels are less important than the cognitive demands inherent to games within those genres.

Another genre of games investigated in the field is the puzzle game genre. Puzzle games are expected to impact areas of cognition that are quite different from those impacted by action (or action-related) games. Puzzle games are generally slower and provide the opportunity for long-range planning. Often, they require participants to overcome cognitive barriers such as functional fixedness (i.e., the tendency to hold a very rigid view of the possible use of objects and thus fail to recognize they could be used in other ways, such as viewing a box as something to store items within and thus failing to recognize that it could be used to stand upon to gain a higher vantage point, for example). Studies in this sphere have utilized a range of puzzle-type games including so-called "AAA" games such as *Portal 2* (i.e., large-budget commercial games from highly successful studios), while other studies have looked at more simplistic games such as mobile or online mini games which often have straightforward, repetitive mechanics and include multiple levels to complete with increasing challenge, sometimes introducing new mechanics as levels progress. Games in this latter category include match-three games such as *Bejeweled* and *Candy Crush*, and time-management games such as restaurant/cooking games like the *Delicious* series or *Diner Dash*. The research on mini games is relatively sparse, but a correlational study by Baniqued et al. (2013) indicates that it is possible to identify mini games that load onto selected cognitive tasks, including working memory and fluid intelligence.

How Is the Interplay Between Cognitive Function and Video Game Play Studied?

While those interested in reading about the relationship between video game play and cognitive function would likely not put research methods on the top of the list of their most anticipated topics, a few key types of studies are critical to understand in order to determine whether the predictions in the field are or are not borne out.

There are two overarching approaches most commonly used in the literature—correlational studies and intervention studies. In correlational studies, researchers recruit participants and measure (at least) two things: (a) each individual's history of video game play, and (b) their level of performance on one or more measures of cognitive function. In short, researchers using this broad type of design are seeking to examine whether individual differences in cognitive function are related to individual differences in video game playing habits.

Within the broad category of correlational studies, there are many more specific designs. In some studies, researchers collect data from a wide range of game player types (e.g., people who play multiple types of games, who play for different amounts of time, etc.). The goal, then, is to assess whether game-playing habits, as operationalized in a more continuous fashion (i.e., those who play 1 hour per week, those who play 2 hours per week, . . . those who play 10 hours per week), are predictive of individual differences in given cognitive skills. In other studies, researchers specifically recruit individuals from the extreme ends of the game-playing spectrum (e.g., one group that rarely plays action video games and one group that very frequently plays action games). Here the goal is much the same, but rather than being continuous, the analysis is more categorical in nature (i.e., whether the group that plays a lot of a given game type differs in cognitive function from the group that plays very little of that game type). In still other cases, rather than using amount of game play time as the critical measure, game skill is assessed instead. Many modern games provide easy-to-collect quantitative markers of skill. Here the analyses seek to examine whether more skillful players outperform less skillful players on the measures of cognitive function.

Correlational studies have several key advantages as well as a number of disadvantages. In terms of advantages, they are reasonably easy and inexpensive to perform and can be used to assess whether a relationship exists between playing (or excelling at playing) certain types of games and differences in cognitive function. They thus provide a window into the cognitive demands inherent in those game types. The main disadvantage, though, is that such studies cannot explain why a relationship exists (if it does). More specifically, if a relation is found between playing a certain type of game and having a higher level of a particular cognitive skill, a correlational design cannot tell you which caused which. It could be the case, as per the main theoretical perspective in the field, that playing the given game type places load on the cognitive function of interest, and that in turn causes the enhancement in cognitive skill. However, it could also be the case that people who naturally have higher levels of cognitive skill are attracted to playing that particular game type (because, due to their higher level of cognitive skill, they are more successful at the game).

The second main type of method—intervention studies, or true experiments—do allow assessments of causality. Generally, in intervention studies, researchers recruit non-video game players (i.e., individuals who do not themselves typically choose to play video games often if at all). They first administer measurements of cognitive skills of interest to the study to all participants before then randomly assigning them to either a control or experimental group. Members of the experimental group are trained on a video game or set of games thought to potentially impact the measured cognitive area. Control participants are engaged in a similar activity to the trained game (e.g., another commercially successful video game), but one that does not include the components hypothesized to make an impact on the cognitive skill, for the duration of the intervention. For example, if the hypotheses are related to demand placed on perceptuo-cognitive skills, the experimental group may be trained on an action video game. Control group members, meanwhile, would be trained on a game that is rated as fun and engaging as the action game, but, critically, that lacks all “action” mechanics (typical control games for these types of studies include slow life-simulation games like *The Sims* or *Restaurant Empire*). After completing training, which can last between a few hour-long sessions and many weeks or months of regular play, participants are tested again on the cognitive area of interest. Importantly, these post-tests always occur at least a full day after the last game play session, so as to ensure that any behavioral results cannot be attributed to transient changes induced by game play (e.g., certain game types will activate the so-called “fight-or-flight” system, which has consequences for a host of human processing abilities). Finally, performance gains are compared between the experimental and control groups. If significant performance improvements seen in participants playing the game that is hypothesized to be related to the measured cognitive skill, but not by the group involved in the control activity, researchers can infer that the improvements were caused by playing the game of interest.

It is thus apparent that the ability to make these causal claims with confidence relies heavily on the quality and efficacy of the study methods and, especially, the appropriateness of the control activity. For example, a study that compares an experimental group playing a highly engaging, fast-paced shooter game with a control group tasked with solving word puzzles on paper for 15 hours is likely not controlling adequately for the many possible confounding effects present in this design that have nothing to do with the cognitive skill of interest (e.g., algorithmic difficulty adjustments based on player performance and other attributes which may produce unequal engagement between video games and control tasks). Indeed, the type of control group employed in an intervention study indicates not only the quality of the study, but also the depth of the inferences that can be drawn from its results. Thus, the research on the impact of video games on cognition utilizes multiple study designs to both investigate possible associations between cognition and games and to establish causal relationships as well.

How Do Different Types of Games Differentially Impact Cognition?

Given the theory underlying why certain types of game play might be linked to cognitive skills and the methods employed to test those theories, results in the field, organized in this section by game type, reflect the typical differences in demands across these types of games.

Action Games

Games that have been categorized as action games were some of the first to be explored by researchers, primarily because they require fast-paced decision-making; thus, players whose cognitive abilities allowed them to complete the necessary actions more quickly would perform better. Researchers realized that players who had dedicated many hours to practicing these skills would likely then excel on tasks measuring skills such as multiple object tracking, visual decision making, and useful field of view (UFOV). Initial studies established that avid players of these types of games outperformed nonplayers in these and other key areas of perceptual cognition through cross-sectional work (e.g., Green & Bavelier, 2003). Subsequent research identified action game play as the causal agent responsible for the observed difference in cognitive skills via several well-controlled intervention studies (Feng et al., 2007; Green & Bavelier, 2007; Wu & Spence, 2013). For instance, in seminal work by Green and Bavelier, participants (who rarely or never naturally chose to play action video games) were trained on the action game *Medal of Honor*. This game, like most first-person shooter action games, requires players to attend to multiple objects or agents in their environment simultaneously, and to alternate quickly between attentional targets for successful performance (Green & Bavelier, 2003). Participants who were trained on this action game over the course of 10 days improved their performance on a number of measures of visual cognition as compared to an active control group that was trained on a non-action game. These measures included assessments of visual search (i.e., to find a particular target in a cluttered scene) and temporal attention (i.e., to parse a quickly changing scene to find certain targets). The results of this, and other similar studies, illustrate that action video game play can enhance aspects of visual attention and visual working memory, resulting in improvements in performance on a variety of related tasks. Later work identified positive impacts of action video games on additional cognitive skills such as attentional control, task switching, and some components of short-term/working memory. For example, studies have indicated that action game players outperform nonplayers on task-switching measures both in correlational studies and after explicit action game interventions (Green et al., 2012; Karle et al., 2010). In addition to the direct benefits of these skills, they also predict important higher-level cognitive skills, including problem-solving, planning, and intelligence. For example, it has been posited that speed of processing skills underlie many of what are considered high-level cognitive skills, such as fluid intelligence. Indeed, research has shown that playing action games that load on speed of processing is also positively correlated with performance on measures of fluid intelligence (Unsworth et al., 2015). This relationship may be mediated by the cognitive load placed on speed of processing by gameplay, as quick processing skills are advantageous for better performance on measures of fluid intelligence. More research is necessary to understand the degree to which speed of processing may be responsible for fluid intelligence scores, and to explore additional cognitive skills that may act as foundational components of fluid intelligence, as well as other higher-order cognitive constructs. The literature on the impacts of action games is the largest and most robust in the field and thus has been subject to several meta-analyses over the years (Bediou et al., 2018, 2023; Powers & Brooks, 2014; Powers et al., 2013; Wang et al., 2016). While in general, meta-analyses of this genre have largely found robust positive effects on various levels and components of cognition, some meta-analytic work, in particular work that treated all video game types as equivalent or that has utilized definitions of action games that are broader than what is typical in the field have found lesser results (e.g., Sala et al., 2018; for a more negative view of the field, see Boot et al., 2011; Hilgard et al., 2019). Thus, research should be careful to control for these constructs when conducting meta-level analyses. Still other genres have been less thoroughly investigated; however, there are multiple lines of research regarding each.

Action-Role Playing Games

Like primarily action games, time spent playing Action-Role Playing Games (A-RPG) has been found to correlate with measurements of visuo-spatial abilities (Dale, Kattner, et al., 2020). As discussed above, the action elements present in games within this hybrid genre make gameplay at times indistinguishable from traditional Role Playing Games and at other times very combat-heavy and thus closely resembling action games. Research by Dale and colleagues on a sample of nearly 300 participants classified into several gamer and non-gamer categories, found that players of modern Role Playing Games outperformed non-gamers on a measure of UFOV, and performed similarly to action video game players on the same task. This indicates that, at least in terms of the load placed on the visual attention skills required for monitoring a wide scope of the visual field, ARPGs provide comparable training to action games.

Real Time Strategy/Multiplayer-Online Battle Arena Games

Despite not containing some mechanics thought to positively impact cognition present in action video games, Real Time Strategy (RTS) games such as *Rise of Nations* and *StarCraft II* have been found to be associated with and improve attentional control/cognitive flexibility, task switching, and speed of processing (Dale & Green, 2017a; Glass et al., 2013; Large et al., 2019). Research on RTS game players has found that avid players outperform non-game players on measures of reaction time, while training non-game players on RTS games for 40 hours resulted in improvements in cognitive flexibility skills (Dale & Green, 2017a; Glass et al., 2013). Further, work by Basak et al. (2008) found that training older adults for less than 24 total hours on the RTS game *Rise of Nations* improved performance on measures of task switching, working memory, visual short-term memory, and mental rotation. This body of work supports the hypothesis that some common mechanisms between action and RTS games may be responsible for improvements both in perceptual and cognitive skills.

A popular subgenre of RTS games are multiplayer-online battle arena games. This subgenre includes the likes of *League of Legends* and *Defense of the Ancients* 2 (*Dota* 2). Research investigating the impacts of *League of Legends* found that player rankings were positively associated with speed of processing and attentional abilities in a correlational study (Large et al., 2019). In a similar study, players of *League of Legends* with higher rankings also performed better on a measure of cognitive flexibility (Valls-Serrano et al., 2022). Research on *Dota* 2 has indicated that higher-ranking players were better able to make decisions under ambiguity and make spatial decisions more quickly than those with lower rankings (Bonny et al., 2016; Sörman et al., 2022).

Puzzle Games

Playing games with strategy and puzzle mechanics has been found to improve problem-solving skills. In a review of the impact of games on selected high-level cognitive skills, Parong and colleagues found that puzzle games were the most likely game type to produce improvements in problem-solving skills when compared to other game types or passive control activities (Parong et al., 2021). It is unclear how far this trend will extend due to the heterogeneity of the genre of puzzle games, which span from massive, commercially developed AAA games such as *Portal* 2 to small, free, online level-based mini games. However, the nature of problem-solving is such that while the ability to utilize well-known paths to reach a goal, such as formulas in mathematics, are readily taught in educational settings, the ability to creatively problem-solve in new contexts where no personally known paths exist requires strong and flexible cognitive problem-solving skills, which can only be developed through practice. Thus, games that require players to employ previously unknown strategies or adjust known routes toward a solution of a puzzle may improve the flexibility with which they are able to reassess assumptions and navigate problem spaces in general. For example, Shute et al. (2015) found that 8 hours of playing *Portal* 2, a puzzle-platforming game where players must create interconnecting portals to solve each puzzle, improved performance on multiple types of problem-solving tasks more than playing cognitive training games.

From this review of the literature on the impact of video games of all types on cognitive functioning, it is apparent that playing games does not, in fact, "rot" players' brains as has been purveyed by popular culture. However, despite the research to the contrary, this belief still permeates (e.g., news stories still report in a surprised manner when the results suggest that the opposite is true; see Mann, 2022). Nonetheless, the field will continue to build upon itself and address the current gaps to build a stronger and clearer understanding of the vast array of impacts games at large may have on cognition.

Where Should the Field Go From Here?

Diversifying Studied Game Genres

Despite broad interest in the impact of video games on cognition for the better part of half a century and the exponential increase in research production on this topic since the early 2000s, only a fraction of the existing game space has been explored. The bulk of the work in this field has focused on action games, and although they have been shown to have important impacts on cognitive skills, especially lower-level cognitive skills, many other interesting avenues of research are underexplored in comparison. Future research questions of interest might range from deciphering the impacts of other game types (e.g., strategy games, puzzle games, and mobile games, to name a few) on cognition, to how games can be used to measure or train latent cognitive constructs such as those related to educational success. Indeed, it is important to differentiate between the myriad game types that may have varied impacts on cognitive skills and the areas of cognition that they are theorized to impact based on the mechanics they employ. Different games differentially impact cognitive areas from low-level cognitive skills such as perceptual abilities, to high-level skills such as fluid intelligence and creativity. This range of skills must be addressed in studies appropriate to the requirements for each skill in studies on games. A single game may even impact multiple cognitive skills in different strata of cognition via diverse mechanisms included in the gameplay experience or environment. This is even more relevant because games are increasingly allowing a wider range of possible strategies (e.g., some players may choose to navigate a game primarily using stealth, while others may opt for heavy direct combat in their playthrough) and player-selected character classes (e.g., soldier-type characters have traits that often promote the use of brute-force combat, while engineer-type characters have traits that encourage players to focus on taking out enemies though remote mine detonation or turret construction, as in *Team Fortress* 2). This means that players may have vastly different experiences within the same games and therefore receive different amounts of training on even what might be considered the central mechanics of a particular game. These nuances demand further investigation to understand the unique impacts of the hybridization of game mechanics and differing player styles and experiences on cognition.

Research to date has focused primarily on AAA games and their players despite enormous public engagement with online and mobile mini games and puzzle games. For instance, the popular match-three puzzle game *Candy Crush* had 255 million active users in 2021, while the mobile strategy game *Clash of Clans* had 76 million users in 2021 (Curry, 2023; Dhillon, 2023). It is thus pertinent to understand the potential impact of these and other mobile and mini games on cognition due to their prolific usage by the public. Additionally, the relative simplicity of mini games lends itself to adaptation as measurement tasks for related cognitive skills. In general, they employ few in-game mechanics but maintain the ecological validity and engagement of larger games where traditional cognitive measurement tasks may lack these qualities. Work such as Baniqued et al. (2013) indicates that it is possible to identify a set of mini games that load onto selected cognitive tasks, and therefore the cognitive faculties required to succeed at those tasks, including working memory and fluid intelligence. And Oei and Patterson (2013) present evidence that mini games may benefit multiple areas of cognition via a training study utilizing several mobile mini games. Furthering this line of work, research can identify the cognitive skills that are loaded on by different types of mini games in order to explore the possible cognitive impacts of said games and/or their usability as measurement paradigms as well as to examine how the many differences between mini games and AAA games alter their impact (e.g., mini games are often played on smaller screens; this may reduce the extent to which they load on peripheral processing/distributed attention but may introduce additional crowding/need for selective attention).

Diversifying Studied Cognitive Abilities

In addition to the breadth of the game space, the cognitive functions tapped by gameplay are not yet fully understood with much specificity. While it is relatively certain that visual attention and cognitive flexibility, for example, are indeed improved through several types of gameplay, it is yet unclear whether lesser studied cognitive areas are equally modifiable through similar means, or modifiable at all. For instance, there is reason to believe that bottom-up attention, which is subcortically driven, might be less malleable by outside influences. Thus, no matter how much cognitive load is placed upon it, it may remain unaltered. On the other hand, regarding observed impacts on higher cognitive skills, the open question remains of whether it is the higher cognitive skill itself (e.g., problem solving), or the lower-level cognitive capacities that underlie that skill (e.g., working memory or inhibitory control) to which performance on measures of the high-level skill can be primarily attributed. Further research is needed to understand the foundational cognitive skill components of high-level cognitive skills, since these skills present as latent attributes arising from the interworking of many lower-level cognitive skills and thus do not have direct or even indirect neural correlates. Understanding the components of cognition that form the foundation of abstract cognitive capacities such as problem-solving and intelligence may allow researchers to more accurately measure and more effectively improve these capabilities.

Distinguishing Between Skill Transfer and Learning to Learn

Another application of the work growing from this field that would benefit from deeper exploration is the differentiation between simple immediate transfer between training and measurement tasks and a skill that has been called "learning to learn." Direct cognitive skill transfer applications include simulations which allow prospective pilots to train and practice their skills without requiring aircraft (Mckinley et al., 2011), and technology-based interventions which provide reading training without directly requiring reading (Franceschini et al., 2015; Pasqualotto, Altarelli, et al., 2022). However, in some cases it can be argued that training improvements seen in a posttest that takes place within a day or two after the training activity may be the result of activation of pertinent areas of the brain during training that are thus more activated during posttest. If this is the case, most if not all the benefits seen from the training will be lost after some time when activation returns to baseline. Therefore, some of the improvements seen after cognitive training may be due in part to the strengthening of mechanisms that allow for efficient development of new cognitive skills rather than direct training of the skills of interest. This is known as learning to learn. This type of training, however, is likely targeting a different type of cognitive skill from direct transfer of skill learning and thus would likely rely on different mechanisms in the brain and require different methods and theories to investigate effective paradigms for improvement. This distinction opens the possibility for deeper exploration of the mechanisms responsible for observed performance enhancements from video games in all genres. Indeed, the current methods utilized in the field may not adequately capture the possible differences between the impacts of direct skill transfer and learning to learn (for a discussion of the theoretical bases for learning to learn in the field see Zhang et al., 2021).

Exploring Individual-Level Predictors of Differences in Training Impact

Furthermore, intuitively, there are differences in the starting skill level, the rate of skill attainment, and the maximum possible attainment across individuals. This means that improvements obtained from training on action games will have different impacts depending on the individual; however, most if not all individuals will make some improvements over time. Another moderator of the impacts of training on cognitive skills is motivation and subsequent engagement with the training activity. Theories in the field endeavoring to explain the effectiveness of action games as training paradigms posit that a key component is the fact that games are fun. While this may seem reductive, it follows that if individuals are highly intrinsically motivated to dedicate their attention and cognitive resources for long periods of time to a particular activity requiring active engagement of their cognitive faculties, this would have positive implications for their cognition. Thus, individual differences in motivation and engagement with games may impact the effectiveness of training outcomes as well.

In sum, future research should seek to address several gaps in the current understanding of how training of cognitive skills can be accomplished via video game play. First, the breadth of game genres that have been explored in the literature lacks emphasis on the diverse experiences that can now be attained in modern games and on smaller games such as mini games and level-based puzzle games, despite their incredible popularity. Second, more research is needed to adequately understand the effects of games on higher-level cognitive skills such as problem-solving and planning skills. Third, future research should address the differences between training for the purpose of direct transfer of cognitive skills and training for the purpose of training learning to learn. This distinction becomes more pertinent as the field evolves to more fine-grained interventions and measurement methods (see Green et al., 2019 for a discussion of the current methodological standards in the field). Differentiating between the goals of training will allow development of more precise and appropriate theories that may indeed spur previously unaddressed research questions and applications of this work.

Primary Sources

For those interested in a thorough overview of some of the most influential published research within the field, this section provides a roughly chronological summary of that work with a brief description of each. Over time, the popular perception of video games has evolved from mindless or even cognitively detrimental entertainment to their current perception as uniquely applicable activities utilized in many diverse facets of life. To fully understand this transition, seminal work beginning in the 1980s with Griffith et al. (1983) and the 1990s with Greenfield, Brannon, and Lohr (1994) along with other groups, must be highlighted. These groups demonstrated that the activities required to be successful at video games made them ideal for strengthening both visuo-motor and cognitive skills in avid players, thus arguing against the common misconception that games might “rot” players’ brains.

As games increased rapidly in widespread popularity and accessibility due to advancements in at-home console technology in the late 1990s and early 2000s, research in the field began to explore the specific cognitive impacts of the video game genre dubbed “action games” on players of these games, action game players. Green and Bavelier (2003) found that action game players outperformed non-game players on measures of visuospatial attention. In addition to these correlational effects, they extended their findings by training non-game-playing participants on an action game, which resulted in enhanced performance on visual selective attention as well, thus providing causal evidence for the relationship between these games and cognitive improvements. Research expanding these results can be found in papers such as Feng et al. (2007), which found that action video game training improved spatial cognition skills, and Strobach et al. (2012), which found that players of this genre also possessed enhanced executive control and task-switching skills, in addition to further work by Green and Bavelier (2006, 2007). See Bediou et al. (2018) for a comprehensive review and meta-analysis of studies published between 2000 and 2015 that investigate the impacts of action video games on cognitive skills both through correlational studies and intervention studies. For a review of the literature focusing on action video games and spatial cognition, see Spence and Feng (2010). Additionally, Bediou et al. (2023) offers a follow-up and expansion of their previous meta-analysis spanning 20 years of literature between 2000 and 2020.

As the video game industry developed into the 2010s, the previously common occurrence of games exclusively belonging to a specific game genre began to dwindle as hybrid genre games became more popular. These new games included components from multiple genres, thus complicating studies which previously could easily identify “action” and “non-action” game players based on the titles they played. In addition to the changes in common distributions of game mechanics, gamer demographics also began changing to include more women and a wider range of player ages from young children to older adults. Work by Dale and Green (2017b) addressed these two new attributes and the adjustments researchers needed to make to accommodate these changes. In general, less work has been dedicated to games falling outside of the action game genre. Regardless, important research has still been done. For example, Dale, Kattner, et al. (2020) found that avid Action-Role Playing Game players had enhanced attentional processing, while Glass et al. (2013) demonstrated that training on a Real Time Strategy (RTS) game produced greater improvements in cognitive flexibility as compared to a control group trained on a simulation game. In a study with older adults at risk of declining cognitive functioning, Basak et al. (2008) provided evidence that playing a RTS game produces improvements as compared to controls in task switching, working memory, visual short-term memory, and mental rotation skills. Finally, Shute et al. (2015) found that participants trained on a popular puzzle-platform game outperformed controls on measures of problem-solving, while Baniqued et al. (2013) found that performance on individual online mini games correlated with relevant selected cognitive skills.

In the 2010s and 2020s, work in the field turned to focus on ensuring methodological and theoretical rigor is present in future contributions to this body of work, as well as expanding the scope of mechanisms and applications from these contributions. Important standards of methodology and improvements to the status quo are laid out in Green et al. (2019), specifically with respect to the gold standard of intervention study for cognitive enhancement. Additionally, recent work has addressed a growing question about whether some training gains acquired from gameplay may be the result of training on a more metacognitive skill dubbed “learning to learn,” as discussed by Zhang et al. (2021). Finally, Pasqualotto, Parong, et al. (2022) examine the components of effective video game training for educational purposes and make design suggestions for optimal scholastic impact.

Reviews of the Literature

For a comprehensive overview of the literature addressing the effects of video games on cognition, various major reviews and books are suggested. For a meta-analysis of the literature on action video games and cognitive skills generally see Bediou et al. (2018, 2023). See Anguera and Gazzaley (2015) for a review clearly delineating between the impacts of games and those of cognitive tasks. For a review of the effects of action video game interventions on attentional control skills see Bavelier and Green (2019) and see Spence and Feng (2010) for a review of intervention effects on spatial cognition. Unsworth et al. (2015) review broadly the impacts of games on cognition and find little evidence for positive effects, while Green et al. (2017) presents a counter argument to this by distinguishing between game types. Dale, Joessel, et al. (2020) presents an overview of the shifting landscape of video game genres and the resulting need for shifts in methodology within the field of cognitive neuroscience. In books, *Video Games: A Medium That Demands Our Attention* (Bowman, 2020) discusses video games and the work to date studying them, while *Computer Games and Team and Individual Learning* (O'Neil & Perez, 2007) covers theoretical and empirical viewpoints of learning from video game training in domestic and military contexts. For elaboration on the impacts of games for education and learning applications, *Computer Games for Learning: An Evidence-Based Approach* (Mayer, 2014) offers perspectives on the utilization of games for learning educational content and academic skills while *Handbook of Game-Based Learning* (Plass et al., 2020) provides empirically backed suggestions for the implementation of using video games for educational outcomes. Finally, a chapter by Blumberg and Ismailer (2009) in *Serious Games* addresses the impact of video games on learning outcomes in children. Together, these titles offer a breadth and depth of scholarship on the effects of video games on cognition and learning.

Further Reading

Anguera, J. A., & Gazzaley, A. (2015). Video games, cognitive exercises, and the enhancement of cognitive abilities *<https://doi.org/10.1016/j.cobeha.2015.06.002>*. *Current Opinion in Behavioral Sciences*, *4*, 160–165.

Bavelier, D., & Green, C. S. (2019). Enhancing attentional control: Lessons from action video games *<https://doi.org/10.1016/j.neuron.2019.09.031>*. *Neuron*, *104*(1), 147–163.

Bediou, B., Adams, D. M., Mayer, R. E., Tipton, E., Green, C. S., & Bavelier, D. (2018). Meta-analysis of action video game impact on perceptual, attentional, and cognitive skills *<https://doi.org/10.1037/bul0000130>*. *Psychological Bulletin*, *144*, 77–110.

Blumberg, F., & Ismailer, S. (2009). What do children learn from playing digital games? In U. Ritterfeld, M. Cody, & P. Vorderer (Eds.), *Serious games* (pp. 131–142). Routledge.

Bowman, N. D. (2020). *Video games: A medium that demands our attention*. Routledge.

Dale, G., Joessel, A., Bavelier, D., & Green, C. S. (2020). A new look at the cognitive neuroscience of video game play *<https://doi.org/10.1111/nyas.14295>*. *Annals of the New York Academy of Sciences*, *1464*(1), 192–203.

Green, C. S., Kattner, F., Eichenbaum, A., Bediou, B., Adams, D. M., Mayer, R. E., & Bavelier, D. (2017). Playing some video games but not others is related to cognitive abilities: A critique of Unsworth et al. (2015) *<https://doi.org/10.1177/0956797616644837>*. *Psychological Science*, *28*(5), 679–682.

Mayer, R. E. (2014). *Computer games for learning: An evidence-based approach* *<https://doi.org/10.7551/mitpress/9427.001.0001>*. MIT Press.

O'Neil, H., & Perez, R. (2007). *Computer games and team and individual learning*. Elsevier.

Plass, J. L., Mayer, R. E., & Homer, B. D. (2020). *Handbook of game-based learning*. MIT Press.

Spence, I., & Feng, J. (2010). Video games and spatial cognition <https://doi.org/10.1037/a0019491>. *Review of General Psychology*, *14*(2), 92–104.

Unsworth, N., Redick, T. S., McMillan, B. D., Hambrick, D. Z., Kane, M. J., & Engle, R. W. (2015). Is playing video games related to cognitive abilities? <https://doi.org/10.1177/0956797615570367> *Psychological Science*, *26*(6), 759–774.

References

Anguera, J. A., & Gazzaley, A. (2015). Video games, cognitive exercises, and the enhancement of cognitive abilities <https://doi.org/10.1016/j.cobeha.2015.06.002>. *Current Opinion in Behavioral Sciences*, *4*, 160–165.

Baniqued, P. L., Lee, H., Voss, M. W., Basak, C., Cosman, J. D., DeSouza, S., Severson, J., Salthouse, T. A., & Kramer, A. F. (2013). Selling points: What cognitive abilities are tapped by casual video games? <https://doi.org/10.1016/j.actpsy.2012.11.009> *Acta Psychologica*, *142*(1), 74–86.

Basak, C., Boot, W. R., Voss, M. W., & Kramer, A. F. (2008). Can training in a real-time strategy video game attenuate cognitive decline in older adults? *Psychology and Aging*, *23*(4), 765–777.

Bavelier, D., & Green, C. S. (2019). Enhancing attentional control: Lessons from action video games <https://doi.org/10.1016/j.neuron.2019.09.031>. *Neuron*, *104*(1), 147–163.

Bavelier, D., Green, C. S., Pouget, A., & Schrater, P. (2012). Brain plasticity through the life span: Learning to learn and action video games <https://doi.org/10.1146/annurev-neuro-060909-152832>. *Annual Review of Neuroscience*, *35*(1), 391–416.

Bediou, B., Adams, D. M., Mayer, R. E., Tipton, E., Green, C. S., & Bavelier, D. (2018). Meta-analysis of action video game impact on perceptual, attentional, and cognitive skills <https://doi.org/10.1037/bul0000130>. *Psychological Bulletin*, *144*, 77–110.

Bediou, B., Rodgers, M. A., Tipton, E., Mayer, R. E., Green, C. S., & Bavelier, D. (2023, Spring). Effects of action video game play on cognitive skills: A meta-analysis <https://doi.org/10.1037/TMB0000102>. *Technology, Mind, and Behavior*, *4*(1).

Blumberg, F., & Ismailer, S. (2009). What do children learn from playing digital games? In U. Ritterfeld, M. Cody, & P. Vorderer (Eds.), *Serious games* (pp. 131–142). Routledge.

Bonny, J. W., Castaneda, L. M., & Swanson, T. (2016, May). *Using an international gaming tournament to study individual differences in MOBA expertise and cognitive skills* <https://doi.org/10.1145/2858036.2858190> [Conference session]. Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems in San Jose California USA.

Boot, W. R., Blakely, D. P., & Simons, D. J. (2011). Do action video games improve perception and cognition? <https://doi.org/10.3389/fpsyg.2011.00226> *Frontiers in Psychology*, *2*, 226.

Bowman, N. D. (2020). *Video games: A medium that demands our attention*. Routledge.

Curry, D. (2023, January 9). Candy crush revenue and usage statistics (2023) *<https://www.businessofapps.com/data/candy-crush-statistics/>*. *Business of Apps*.

Dale, G., & Green, C. S. (2017a). Associations between avid action and real-time strategy game play and cognitive performance: A pilot study *<https://doi.org/10.1007/s41465-017-0021-8>*. *Journal of Cognitive Enhancement*, *1*(3), 295–317.

Dale, G., & Green, C. S. (2017b). The changing face of video games and video gamers: Future directions in the scientific study of video game play and cognitive performance *<https://doi.org/10.1007/s41465-017-0015-6>*. *Journal of Cognitive Enhancement*, *1*(3), 280–294.

Dale, G., Joessel, A., Bavelier, D., & Green, C. S. (2020). A new look at the cognitive neuroscience of video game play *<https://doi.org/10.1111/nyas.14295>*. *Annals of the New York Academy of Sciences*, *1464*(1), 192–203.

Dale, G., Kattner, F., Bavelier, D., & Green, C. S. (2020). Cognitive abilities of action video game and role-playing video game players: Data from a massive open online course *<https://doi.org/10.1037/ppm0000237>*. *Psychology of Popular Media*, *9*(3), 347–358.

Dhillon, S. (2023, June 6). Clash of clans revenue and usage statistics (2023) *<https://www.businessofapps.com/data/clash-of-clans-statistics/>*. *Business of Apps*.

Dorval, M., & Pépin, M. (1986). Effect of playing a video game on a measure of spatial visualization *<https://doi.org/10.2466/pms.1986.62.1.159>*. *Perceptual and Motor Skills*, *62*, 159–162.

Feng, J., Spence, I., & Pratt, J. (2007). Playing an action video game reduces gender differences in spatial cognition *<https://doi.org/10.1111/j.1467-9280.2007.01990.x>*. *Psychological Science*, *18*(10), 850–855.

Franceschini, S., Bertoni, S., Ronconi, L., Molteni, M., Gori, S., & Facoetti, A. (2015). “Shall we play a game?” Improving reading through action video games in developmental dyslexia *<https://doi.org/10.1007/s40474-015-0064-4>*. *Current Developmental Disorders Reports*, *2*(4), 318–329.

Glass, B. D., Maddox, W. T., & Love, B. C. (2013). Real-time strategy game training: Emergence of a cognitive flexibility trait *<https://doi.org/10.1371/journal.pone.0070350>*. *PLOS ONE*, *8*(8), e70350.

Green, C. S., & Bavelier, D. (2003). Action video game modifies visual selective attention *<https://doi.org/10.1038/nature01647>*. *Nature*, *423*(6939), 534–537.

Green, C. S., & Bavelier, D. (2006). Enumeration versus multiple object tracking: The case of action video game players *<https://doi.org/10.1016/j.cognition.2005.10.004>*. *Cognition*, *101*(1), 217–245.

Green, C. S., & Bavelier, D. (2007). Action-video-game experience alters the spatial resolution of vision *<https://doi.org/10.1111/j.1467-9280.2007.01853.x>*. *Psychological Science*, *18*(1), 88–94.

Green, C. S., Bavelier, D., Kramer, A. F., Vinogradov, S., Ansorge, U., Ball, K. K., Bingel, U., Chein, J. M., Colzato, L. S., Edwards, J. D., Facoetti, A., Gazzaley, A., Gathercole, S. E., Ghisletta, P., Gori, S., Granic, I., Hillman, C. H., Hommel, B., Jaeggi, S. M., . . . Witt, C. M. (2019). Improving methodological standards in behavioral interventions for cognitive enhancement *<https://doi.org/10.1007/s41465-018-0115-y>*. *Journal of Cognitive Enhancement*, *3*(1), 2–29.

Green, C. S., Kattner, F., Eichenbaum, A., Bediou, B., Adams, D. M., Mayer, R. E., & Bavelier, D. (2017). Playing some video games but not others is related to cognitive abilities: A critique of Unsworth et al. (2015) <https://doi.org/10.1177/0956797616644837>. *Psychological Science*, *28*(5), 679–682.

Green, C. S., Sugarman, M. A., Medford, K., Klobusicky, E., & Bavelier, D. (2012). The effect of action video game experience on task-switching <https://doi.org/10.1016/j.chb.2011.12.020>. *Computers in Human Behavior*, *28*(3), 984–994.

Greenfield, P. M., Brannon, C., & Lohr, D. (1994). Two-dimensional representation of movement through three-dimensional space: The role of video game expertise <https://doi.org/10.1016/0193-3973(94)90007-8>. *Journal of Applied Developmental Psychology*, *15*(1), 87–103.

Greenfield, P. M., DeWinstanley, P., Kilpatrick, H., & Kaye, D. (1994). Action video games and informal education: Effects on strategies for dividing visual attention <https://doi.org/10.1016/0193-3973(94)90008-6>. *Journal of Applied Developmental Psychology*, *15*(1), 105–123.

Griffith, J. L., Voloschin, P., Gibb, G. D., & Bailey, J. R. (1983). Differences in eye-hand motor coordination of video-game users and non-users. *Perceptual and Motor Skills*, *57*(1), 155–158.

Hilgard, J., Sala, G., Boot, W. R., & Simons, D. J. (2019). Overestimation of action-game training effects: Publication bias and salami slicing <https://doi.org/10.1525/collabra.231>. *Collabra: Psychology*, *5*(1), 30.

Karle, J. W., Watter, S., & Shedden, J. M. (2010). Task switching in video game players: Benefits of selective attention but not resistance to proactive interference <https://doi.org/10.1016/j.actpsy.2009.12.007>. *Acta Psychologica*, *134*(1), 70–78.

Large, A. M., Bediou, B., Cekic, S., Hart, Y., Bavelier, D., & Green, C. S. (2019). Cognitive and behavioral correlates of achievement in a complex multi-player video game <https://doi.org/10.17645/mac.v7i4.2314>. *Media and Communication*, *7*(4), 198–212.

Mann, R. (2022, October 26). Are video games bad for children? New UVM research says maybe not <https://www.wcax.com/2022/10/26/are-video-games-bad-children-it-seems-new-research-says-maybe-not/>. *WCAX*.

Mckinley, R., McIntire, L., & Funke, M. (2011). Operator selection for unmanned aerial systems: Comparing video game players and pilots <https://doi.org/10.3357/ASEM.2958.2011>. *Aviation, Space, and Environmental Medicine*, *82*, 635–642.

Mayer, R. E. (2014). *Computer games for learning: An evidence-based approach* <https://doi.org/10.7551/mitpress/9427.001.0001>. MIT Press.

Oei, A. C., & Patterson, M. D. (2013). Enhancing cognition with video games: A multiple game training study <https://doi.org/10.1371/journal.pone.0058546>. *PLOS ONE*, *8*(3), e58546.

O’Neil, H., & Perez, R. (2007). *Computer games and team and individual learning*. Elsevier.

Parong, J., Holman, C., Cunningham, E., Green, C. S., & Bavelier, D. (2021). Video games and higher cognition. In Harold F. O'Neil, Eva L. Baker, Ray S. Perez, & Stephen E. Watson (Eds.), *Using cognitive and affective metrics in educational simulations and games* (pp. 3–30). Routledge.

Pasqualotto, A., Altarelli, I., De Angeli, A., Menestrina, Z., Bavelier, D., & Venuti, P. (2022). Enhancing reading skills through a video game mixing action mechanics and cognitive training *<https://doi.org/10.1038/s41562-021-01254-x>*. *Nature Human Behaviour*, *6*(4), 545–554.

Pasqualotto, A., Parong, J., Green, C. S., & Bavelier, D. (2022). Video game design for learning to learn. *International Journal of Human–Computer Interaction*, *39*(11), 2211–2228.

Plass, J. L., Mayer, R. E., & Homer, B. D. (2020). *Handbook of game-based learning*. MIT Press.

Powers, K., & Brooks, P. (2014). *Evaluating the specificity of effects of video game training <https://doi.org/10.1093/acprof:osobl/9780199896646.003.0021>* In Fran C. Blumberg (Ed.), *Learning by playing: Video gaming in education* (pp. 302–329). Oxford Academic.

Powers, K. L., Brooks, P. J., Aldrich, N. J., Palladino, M. A., & Alfieri, L. (2013). *Effects of video-game play on information processing: A meta-analytic investigation*. Psychonomic Bulletin & Review, *20*, 1055–1079.

Sala, G., Tatlidil, K. S., & Gobet, F. (2018). Video game training does not enhance cognitive ability: A comprehensive meta-analytic investigation *<https://doi.org/10.1037/bul0000139>*. *Psychological Bulletin*, *144*, 111–139.

Shute, V. J., Ventura, M., & Ke, F. (2015). The power of play: The effects of Portal 2 and Lumosity on cognitive and noncognitive skills *<https://doi.org/10.1016/j.compedu.2014.08.013>*. *Computers & Education*, *80*, 58–67.

Sörman, D. E., Dahl, K. E., Lindmark, D., Hansson, P., Vega-Mendoza, M., & Körning-Ljungberg, J. (2022). Relationships between Dota 2 expertise and decision-making ability *<https://doi.org/10.1371/journal.pone.0264350>*. *PLOS ONE*, *17*(3), e0264350.

Spence, I., & Feng, J. (2010). Video games and spatial cognition *<https://doi.org/10.1037/a0019491>*. *Review of General Psychology*, *14*(2), 92–104.

Statista. (2023, June 26). Number of video game users worldwide from 2017 to 2027 (in billions) [Graph] *<https://www.statista.com/statistics/748044/number-video-gamers-world/>*. *Statista*.

Strobach, T., Frensch, P. A., & Schubert, T. (2012). Video game practice optimizes executive control skills in dual-task and task switching situations. *Acta Psychologica*, *140*(1), 13–24.

Unsworth, N., Redick, T. S., McMillan, B. D., Hambrick, D. Z., Kane, M. J., & Engle, R. W. (2015). Is playing video games related to cognitive abilities? *<https://doi.org/10.1177/0956797615570367>* *Psychological Science*, *26*(6), 759–774.

Valls-Serrano, C., de Francisco, C., Caballero-López, E., & Caracuel, A. (2022). Cognitive flexibility and decision making predicts expertise in the MOBA Esport, League of Legends *<https://doi.org/10.1177/21582440221142728>*. *SAGE Open*, *12*(4), 21582440221142730.

Vernon, P. A., Nador, S., & Kantor, L. (1985). Reaction times and speed-of-processing: Their relationship to timed and untimed measures of intelligence *<https://doi.org/10.1016/0160-2896(85)90020-0>*. *Intelligence*, *9*(4), 357–374.

Wang, P., Liu, H.-H., Zhu, X.-T., Meng, T., Li, H.-J., & Zuo, X.-N. (2016). Action video game training for healthy adults: A meta-analytic study *<https://www.frontiersin.org/articles/10.3389/fpsyg.2016.00907>*. *Frontiers in Psychology*, *7*, 907.

Wu, S., & Spence, I. (2013). Playing shooter and driving videogames improves top-down guidance in visual search <*https://doi.org/10.3758/s13414-013-0440-2*>. *Attention, Perception, & Psychophysics*, *75*(4), 673–686.

Zhang, R.-Y., Chopin, A., Shibata, K., Lu, Z.-L., Jaeggi, S. M., Buschkuehl, M., Green, C. S., & Bavelier, D. (2021). Action video game play facilitates "learning to learn <*https://doi.org/10.1038/s42003-021-02652-7*>." *Communications Biology*, *4*(1), 1154.

Related Articles

Game Studies

Games Addiction: A Comprehensive Overview